CAREER COMPASS

Designing an AI-Assisted Career Guidance Experience

A portfolio case study on connecting web development, assessments, Make.com automation and Gemini-powered career recommendations.

HTML5 CSS3 JavaScript Make.com Gemini AI Google Sheets

Tushar Thakur

MCA Data Science · Chandigarh University

PROJECT TYPE	INTERFACE	AUTOMATION	OUTPUT
Career Guidance Prototype	Browser-Based Web App	Make.com Scenarios	Career Roadmap + Email
EXPERIENCE	ASSESSMENT	AI INTEGRATION	STORAGE
Student + Admin Views	Interest / Skills / Goals	Google Gemini	Google Sheets

PROJECT STORY

Why I Built Career Compass

Choosing a career can be overwhelming when interests, skills and academic background do not point to one obvious path. I wanted to explore whether a simple website could guide students through structured self-reflection and turn their responses into more useful career-direction suggestions.

Career Compass became a practical project in combining frontend interfaces with low-code backend automation. I built a landing experience, student and admin-facing screens, an assessment flow, and Make.com workflows designed to coordinate data storage, AI analysis and email communication.

CONTEXT

The Problem

Scattered career information Students often encounter generic recommendations without a single place to reflect on their academic context, interests and preferences.	What Career Compass proposes A connected assessment-to-guidance experience: collect student inputs, process them through an automation workflow, and present an AI-assisted roadmap.

KEY CAPABILITIES

What the Application Can Do

Landing page & account forms Introduce the platform, navigate key sections, and submit registration or login information via configured webhooks.	Student dashboard Provide a personalized student-facing interface with navigation, profile cues and roadmap display.
Career assessment Collect details about academics, skills, interests, preferences and future goals.	AI-assisted recommendations Send assessment responses to a Make.com scenario that integrates Gemini for guidance generation.
Workflow-based communication Use the supplied scenarios for welcome and assessment-report email processes.	Admin interface & themes Include a separate admin dashboard prototype and visual theme controls.

EXPERIENCE

From Student Input to Career Direction

The intended experience follows a student from the main website to an assessment and then toward guidance content. The repository documents the workflow with source files, automation exports, architecture diagrams and screenshots. Actual results depend on correctly configured third-party services.

ARCHITECTURE

How Career Compass Is Structured

The application uses HTML, CSS and vanilla JavaScript for its web interfaces. JavaScript sends requests to Make.com webhooks; Make.com scenarios coordinate downstream services such as Google Sheets, Gemini AI and email integrations. Unlike a traditional server project, the uploaded source does not include a standalone Java, Python or Node.js backend.

LAYER	TECHNOLOGY	RESPONSIBILITY
Interface	HTML / CSS / JavaScript	Landing, student, admin and assessment pages
Orchestration	Make.com webhooks + scenarios	Receive requests, route steps and trigger services
Data and AI	Google Sheets + Gemini	Store workflow data and generate guidance
Communication	Make.com email modules	Deliver configured welcome and report emails

EXAMPLE WORKFLOW

Data Flow: Completing an Assessment

STUDENT → ASSESSMENT FORM → WEBHOOK → MAKE.COM → GEMINI / SHEETS → ROADMAP / EMAIL

After a student completes the assessment, the frontend submits answers through an automation endpoint. The assessment scenario is designed to process the data, connect to AI and data services, and deliver guidance or a report. The student interface can display roadmap content when the expected response or saved data is available.

CORE RELATIONSHIPS

Main Screens and Integrations

COMPONENT	CONNECTS TO	ROLE
MainWeb.html / .js	User workflow webhook	Entry point and account interaction
Assessment.html / .js	Assessment workflow webhook	Collect and submit career inputs
Student.html / .js	Browser state / roadmap content	Show student-facing guidance
Admin.html / .js	Dashboard UI elements	Administrative prototype experience

IMPLEMENTATION

The Engineering Behind the Experience

I chose a lightweight frontend rather than adding a conventional application server. This made the project a focused exercise in user flow, asynchronous requests, automation design and integrations. The exported Make.com scenarios preserve the intended process beyond the visible web pages.

BUILDING BLOCK	EXAMPLES	PURPOSE
Frontend files	MainWeb, Student, Admin, Assessment	Separate responsibilities for key screens
Workflow exports	User Workflow.json, Assessment Workflow.json	Record service orchestration logic
External services	Make.com, Google Sheets, Gemini, email	Support storage, content generation and notifications
Project assets	Architecture diagrams, screenshots, PDFs, demo	Document decisions and demonstrate intended behavior

RELIABILITY

Boundaries and Safeguards

REQUESTS	USER ACCESS	AI OUTPUT	INTEGRATIONS
Validate form inputs	Enforce access on a real backend	Treat guidance as advisory	Use secured accounts and webhooks

This is an academic/portfolio prototype, not a production career-counseling service. Current frontend role-routing is not strong authorization; public webhook endpoints and browser-rendered content also need security hardening before live deployment. The repository’s workflow definitions do not, on their own, prove that external accounts and email delivery remain active.

DESIGN CHOICES

What Made This Project Interesting

A student-first path. The assessment has a clear purpose: translate personal background and preferences into career-oriented suggestions. The key design challenge was keeping the experience approachable rather than exposing the complexity of the underlying automation.

Low-code orchestration. Make.com allowed me to explore how webhooks, storage, AI and email can work together without writing a monolithic backend. That accelerated prototyping while also revealing the importance of integration reliability and credential management.

SCOPE

What the Project Includes — and What It Does Not

INCLUDED ✓ Landing page and authentication forms ✓ Student dashboard interface ✓ Career assessment form ✓ Gemini-enabled automation design ✓ Google Sheets integration design ✓ Welcome and report email workflows ✓ Admin dashboard prototype ✓ Diagrams and project documentation	NOT YET PRODUCTION-READY – Secure server-side authentication – Verified role-based permissions – Fully operational admin management – Guaranteed live third-party connections – Formal recommendation accuracy evaluation – Independent professional counseling – Production monitoring and deployment

REFLECTION

What I Learned

Career Compass taught me how much coordination is required behind a seemingly simple web form. A useful experience depends not only on a polished interface but also on request handling, external-service configuration, data consistency and clear feedback when operations fail.

FRONTEND	AUTOMATION	AI & DATA
Multi-page UI and form interactions	Webhooks, routing and service sequencing	Prompt-driven guidance and Sheets records
SYSTEM DESIGN	RELIABILITY	PROJECT DELIVERY
Workflow / deployment diagrams	Security limitations and failure points	README, screenshots, exports and demo

NEXT ITERATION

What I Would Improve

The most valuable next step would be introducing a secure backend for authentication, authorization and server-side validation. I would then test every workflow end to end, implement robust error handling, separate private configuration, improve the admin tools and evaluate whether the AI recommendations are genuinely useful to students.

CONCLUSION

Final Thoughts

Career Compass started with a question: could a student share interests and skills in one place and receive more structured direction? The resulting prototype demonstrates one approach by joining frontend pages, an assessment, cloud automation, AI-generated recommendations and communication workflows.

For me, its strongest contribution is the system-level learning. I worked across interface design, data capture, workflow exports and service integration, and learned to distinguish an engaging prototype from the security, verification and reliability required for a public product. That distinction is what makes Career Compass an important portfolio project.

EXPLORE

Project Links

GITHUB	https://github.com/tushar-011
LINKEDIN	https://www.linkedin.com/in/tushar-thakur-8848a7396/
EMAIL	artificial.thakur@gmail.com

Career Compass · AI-Assisted Career Guidance Prototype